UNIT 3 · TOPIC 7(A)

Threshers — Capacity, Efficiency and Losses

Syllabus topic: Calculation of capacity, efficiency and losses in threshers, harvesters and chaff cutters

Definitions (per test codes)

- **Total grain input** = grain from ALL outlets.

$$\text{Threshing efficiency} = 100 - \%\ \text{unthreshed grain (unthreshed in straw + main outlet + sieve overflow)}$$

$$\text{Cleaning efficiency} = \text{clean grain at main outlet} \div \text{total material received at main outlet} \times 100$$

- **Losses** (% of total grain): unthreshed, blown/chaffed (grain carried with bhusa), spilled/scattered, sieve overflow, and visible broken grain %.
- **Capacity** = grain output (kg/h) at the loss limits.
- **Specific energy** = kWh per tonne (or quintal) of grain.

Worked example. Total grain 1000 kg/h; unthreshed 10 kg → threshing efficiency = **99%**. The main outlet delivers 950 kg of material, of which 931 kg is clean grain → cleaning efficiency = **98%**. Broken grain 15 kg → **1.5% breakage**. The motor draws 5 kW at 500 kg/h → specific energy = **10 kWh/t = 1 kWh/q** (1 t = 10 q).

Interpreting the results

Efficiency falls and losses climb as the feed rate rises past the optimum. The quoted capacity is at the **maximum feed rate that still meets the loss/breakage limits** (e.g., broken grain within ~2% for seed use; more is tolerated for feed).

Practice MCQs

Q78. Capacity of threshers and combines is expressed on a:

(a) Area basis only
(b) Material (throughput) basis — kg/h grain or kg/s MOG
(c) Fuel basis
(d) Width basis

Q88. Threshing efficiency =

(a) 100 - % broken grain
(b) 100 - % unthreshed grain
(c) Clean grain/total material
(d) Grain/straw ratio

Q89. With 10 kg unthreshed out of 1000 kg total grain, threshing efficiency is:

(a) 90%
(b) 95%
(c) 99%
(d) 100%

Q90. Cleaning efficiency is computed at the:

(a) Straw outlet
(b) Main grain outlet
(c) Feeding chute
(d) Blower

Q91. Main outlet delivers 950 kg material with 931 kg clean grain. Cleaning efficiency ≈

(a) 93%
(b) 95%
(c) 98%
(d) 99%

Q92. A thresher using 5 kW at 500 kg grain/h has specific energy:

(a) 0.1 kWh/q
(b) 1 kWh/q
(c) 10 kWh/q
(d) 100 kWh/q

Q93. Thresher capacity is reported at:

(a) Any feed rate
(b) Maximum feed rate within specified loss/breakage limits
(c) Zero feed
(d) Half speed

Q94. Grain blown out with bhusa constitutes the:

(a) Sieve loss
(b) Blowing/chaff loss
(c) Spilled loss
(d) Cylinder loss

Answers

Q78 — (b) Material (throughput) basis — kg/h grain or kg/s MOG
Q88 — (b) 100 - % unthreshed grain
Q89 — (c) 99% — *(1000-10)/1000 = 99%.*
Q90 — (b) Main grain outlet
Q91 — (c) 98% — *931/950 = 98%.*
Q92 — (b) 1 kWh/q — *5 kW ÷ 0.5 t/h = 10 kWh/t = 1 kWh/q (1 t = 10 q).*
Q93 — (b) Maximum feed rate within specified loss/breakage limits
Q94 — (b) Blowing/chaff loss

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 2 (Field Capacities) & Ch. 17; RNAM test codes (combine, thresher, transplanter); BIS standards for power threshers and chaff cutters; Jagdishwar Sahay; NPTEL lectures.